DATABASE SCHEMA

Smart Emergency Response & Healthcare Coordination Platform

Purpose

Proposed relational database for patient medical information, emergency intake, AI-assisted triage, operator review, hospital coordination, ambulance dispatch and tracking, notifications, and treatment records.

Important

This is a design draft for System Analysis & Design. Validate fields, nullability, key types, and business rules with the team before creating EF Core entities/migrations. All foreign keys referencing AspNetUsers.Id must use the exact same type as the Identity key. No blood-bank functionality is included.

Design principles

- AI triage is decision support; the Emergency Center Operator reviews/confirms the case.
- The hospital accepts or rejects the request and sends its own ambulance with a paramedic.
- Keep status and GPS histories for traceability.
- Protect medical data with authenticated, role-based authorization.

1. Database Tables & Data Dictionary

PK = Primary Key; FK = Foreign Key; UNIQUE = unique constraint; NULL = optional. SQL types are suggested SQL Server types.

1. Patients

Column	SQL type	Key / constraints
PatientID	int / uniqueidentifier	PK
UserID	nvarchar	FK AspNetUsers.Id; UNIQUE
DateOfBirth	date	NULL
Gender	nvarchar(20)	NULL
Address	nvarchar(300)	NULL
CreatedAt	datetime2	NOT NULL

2. MedicalProfiles

Column	SQL type	Key / constraints
MedicalProfileID	int	PK
PatientID	same as Patients.PatientID	FK; UNIQUE
BloodType	nvarchar(5)	NULL
PrimaryConditions	nvarchar(max)	NULL
CurrentNotes	nvarchar(max)	NULL
LastUpdatedAt	datetime2	NOT NULL

3. EmergencyContacts

Column	SQL type	Key / constraints
EmergencyContactID	int	PK
PatientID	same as Patients.PatientID	FK
FullName	nvarchar(150)	NOT NULL
Relationship	nvarchar(50)	NULL
PhoneNumber	nvarchar(30)	NOT NULL
IsPrimary	bit	NOT NULL

4. ChronicDiseases

Column	SQL type	Key / constraints
ChronicDiseaseID	int	PK
Name	nvarchar(150)	UNIQUE, NOT NULL
Description	nvarchar(max)	NULL

5. PatientChronicDiseases

Column	SQL type	Key / constraints
PatientID	same as Patients.PatientID	PK/FK
ChronicDiseaseID	int	PK/FK
DiagnosedAt	date	NULL
Notes	nvarchar(max)	NULL

6. Allergies

Column	SQL type	Key / constraints
AllergyID	int	PK
Name	nvarchar(150)	UNIQUE, NOT NULL
Description	nvarchar(max)	NULL

7. PatientAllergies

Column	SQL type	Key / constraints
PatientID	same as Patients.PatientID	PK/FK
AllergyID	int	PK/FK
Severity	nvarchar(30)	NULL
Reaction	nvarchar(250)	NULL

8. Medications

Column	SQL type	Key / constraints
MedicationID	int	PK
Name	nvarchar(150)	NOT NULL
Description	nvarchar(max)	NULL

9. PatientMedications

Column	SQL type	Key / constraints
PatientID	same as Patients.PatientID	PK/FK
MedicationID	int	PK/FK
Dosage	nvarchar(100)	NULL
Frequency	nvarchar(100)	NULL
StartDate	date	NULL
EndDate	date	NULL

10. EmergencyTypes

Column	SQL type	Key / constraints
EmergencyTypeID	int	PK
Name	nvarchar(100)	UNIQUE, NOT NULL
Description	nvarchar(max)	NULL
DefaultPriority	nvarchar(30)	NULL

11. EmergencyRequests

Column	SQL type	Key / constraints
EmergencyRequestID	bigint	PK
PatientID	same as Patients.PatientID	FK
EmergencyTypeID	int	FK; NULL
Description	nvarchar(max)	NULL
Latitude	decimal(9,6)	NULL (empty when GPS is off)
Longitude	decimal(9,6)	NULL (empty when GPS is off)
AddressText	nvarchar(300)	NULL
Priority	nvarchar(30)	NULL
CurrentStatus	nvarchar(40)	NOT NULL
CreatedAt	datetime2	NOT NULL
PickedUpAt	datetime2	NULL
ArrivedAtHospitalAt	datetime2	NULL
ClosureRequestedAt	datetime2	NULL
ClosedAt	datetime2	NULL
ClosedByUserID	nvarchar(450)	FK AspNetUsers.Id; NULL (hospital staff)

12. AITriageResults

Column	SQL type	Key / constraints
AITriageResultID	bigint	PK
EmergencyRequestID	bigint	FK EmergencyRequests
SuggestedEmergencyTypeID	int	FK EmergencyTypes; NULL
SuggestedPriority	nvarchar(30)	NULL
SuggestedDepartmentID	int	FK Departments; NULL
ConfidenceScore	decimal(5,4)	NULL; 0..1
ReasoningSummary	nvarchar(max)	NULL
CreatedAt	datetime2	NOT NULL

13. EmergencyReviewDecisions

Column	SQL type	Key / constraints
ReviewDecisionID	bigint	PK
EmergencyRequestID	bigint	FK EmergencyRequests
OperatorID	nvarchar(450)	FK AspNetUsers.Id
ConfirmedEmergencyTypeID	int	FK EmergencyTypes; NULL
ConfirmedPriority	nvarchar(30)	NULL
Decision	nvarchar(30)	NOT NULL
Notes	nvarchar(max)	NULL
ReviewedAt	datetime2	NOT NULL

14. CaseStatusHistory

Column	SQL type	Key / constraints
CaseStatusHistoryID	bigint	PK
EmergencyRequestID	bigint	FK EmergencyRequests
Status	nvarchar(40)	NOT NULL
ChangedByUserID	nvarchar(450)	FK AspNetUsers.Id; NULL
Notes	nvarchar(max)	NULL
ChangedAt	datetime2	NOT NULL

15. Hospitals

Column	SQL type	Key / constraints
HospitalID	int	PK
Name	nvarchar(200)	NOT NULL
Address	nvarchar(300)	NOT NULL
Latitude	decimal(9,6)	NULL
Longitude	decimal(9,6)	NULL
PhoneNumber	nvarchar(30)	NULL
IsActive	bit	NOT NULL

16. Departments

Column	SQL type	Key / constraints
DepartmentID	int	PK
HospitalID	int	FK Hospitals
Name	nvarchar(150)	NOT NULL
Description	nvarchar(max)	NULL
Capacity	int	NULL
IsActive	bit	NOT NULL

17. Doctors

Column	SQL type	Key / constraints
DoctorID	int	PK
UserID	nvarchar(450)	FK AspNetUsers.Id; UNIQUE
DepartmentID	int	FK Departments
FullName	nvarchar(150)	NOT NULL
Specialization	nvarchar(150)	NULL
PhoneNumber	nvarchar(30)	NULL
IsActive	bit	NOT NULL

18. HospitalBeds

Column	SQL type	Key / constraints
BedID	int	PK
DepartmentID	int	FK Departments
BedNumber	nvarchar(30)	NOT NULL
BedStatus	nvarchar(30)	Available/Occupied/OutOfService

19. EmergencyHospitalAssignments

Column	SQL type	Key / constraints
EmergencyHospitalAssignmentID	bigint	PK
EmergencyRequestID	bigint	FK EmergencyRequests
HospitalID	int	FK Hospitals
AssignedByOperatorID	nvarchar(450)	FK AspNetUsers.Id; NULL
AssignmentStatus	nvarchar(30)	NOT NULL
AssignedAt	datetime2	NOT NULL
RespondedAt	datetime2	NULL
ResponseNotes	nvarchar(max)	NULL

20. CaseDepartmentAssignments

Column	SQL type	Key / constraints
CaseDepartmentAssignmentID	bigint	PK
EmergencyRequestID	bigint	FK EmergencyRequests
DepartmentID	int	FK Departments
AssignedByUserID	nvarchar(450)	FK AspNetUsers.Id; NULL
AssignedAt	datetime2	NOT NULL
AssignmentStatus	nvarchar(30)	NOT NULL

21. CaseDoctorAssignments

Column	SQL type	Key / constraints
CaseDoctorAssignmentID	bigint	PK
EmergencyRequestID	bigint	FK EmergencyRequests
DoctorID	int	FK Doctors
AssignedByUserID	nvarchar(450)	FK AspNetUsers.Id; NULL
AssignedAt	datetime2	NOT NULL
AssignmentStatus	nvarchar(30)	NOT NULL

22. HospitalCaseAssessments

Column	SQL type	Key / constraints
AssessmentID	bigint	PK
EmergencyRequestID	bigint	FK EmergencyRequests
HospitalID	int	FK Hospitals
AssessedByDoctorID	int	FK Doctors
AssessedPriority	nvarchar(30)	NULL
AssessmentNotes	nvarchar(max)	NULL
AssessedAt	datetime2	NOT NULL

23. Ambulances

Column	SQL type	Key / constraints
AmbulanceID	int	PK
HospitalID	int	FK Hospitals
VehicleNumber	nvarchar(50)	UNIQUE, NOT NULL
AmbulanceStatus	nvarchar(30)	NOT NULL
LastKnownLatitude	decimal(9,6)	NULL
LastKnownLongitude	decimal(9,6)	NULL

24. AmbulanceAssignments

Column	SQL type	Key / constraints
AssignmentID	bigint	PK
EmergencyRequestID	bigint	FK EmergencyRequests
AmbulanceID	int	FK Ambulances
ParamedicID	int	FK Paramedics
AssignedByUserID	nvarchar(450)	FK AspNetUsers.Id; NULL (hospital staff)
AssignedAt	datetime2	NOT NULL
CompletedAt	datetime2	NULL
AssignmentStatus	nvarchar(30)	NOT NULL

25. AmbulanceLocationHistory

Column	SQL type	Key / constraints
LocationHistoryID	bigint	PK
AmbulanceID	int	FK Ambulances
EmergencyRequestID	bigint	FK EmergencyRequests; NULL
Latitude	decimal(9,6)	NOT NULL
Longitude	decimal(9,6)	NOT NULL
RecordedAt	datetime2	NOT NULL

26. AmbulanceStatusHistory

Column	SQL type	Key / constraints
AmbulanceStatusHistoryID	bigint	PK
AmbulanceID	int	FK Ambulances
Status	nvarchar(30)	NOT NULL
ChangedAt	datetime2	NOT NULL
Notes	nvarchar(max)	NULL

27. Notifications

Column	SQL type	Key / constraints
NotificationID	bigint	PK
RecipientUserID	nvarchar(450)	FK AspNetUsers.Id; NULL
EmergencyContactID	int	FK EmergencyContacts; NULL
EmergencyRequestID	bigint	FK EmergencyRequests; NULL
Title	nvarchar(200)	NOT NULL
Message	nvarchar(max)	NOT NULL
Channel	nvarchar(30)	NOT NULL
IsRead	bit	NOT NULL
CreatedAt	datetime2	NOT NULL
SentAt	datetime2	NULL

Each notification goes to a user (RecipientUserID) or to an emergency contact (EmergencyContactID), so one of them must have a value.

28. DoctorNotes

Column	SQL type	Key / constraints
DoctorNoteID	bigint	PK
EmergencyRequestID	bigint	FK EmergencyRequests
DoctorID	int	FK Doctors
NoteText	nvarchar(max)	NOT NULL
CreatedAt	datetime2	NOT NULL

29. PatientTreatmentHistory

Column	SQL type	Key / constraints
TreatmentHistoryID	bigint	PK
EmergencyRequestID	bigint	FK EmergencyRequests
PatientID	same as Patients.PatientID	FK
DoctorID	int	FK Doctors; NULL
TreatmentDetails	nvarchar(max)	NOT NULL
RecordedAt	datetime2	NOT NULL

30. Paramedics

Column	SQL type	Key / constraints
ParamedicID	int	PK
UserID	nvarchar(450)	FK AspNetUsers.Id; UNIQUE
HospitalID	int	FK Hospitals
FullName	nvarchar(150)	NOT NULL
PhoneNumber	nvarchar(30)	NULL
IsAvailable	bit	NOT NULL
IsActive	bit	NOT NULL

31. HospitalStaff

Column	SQL type	Key / constraints
HospitalStaffID	int	PK
UserID	nvarchar(450)	FK AspNetUsers.Id; UNIQUE
HospitalID	int	FK Hospitals
FullName	nvarchar(150)	NOT NULL
PhoneNumber	nvarchar(30)	NULL
IsActive	bit	NOT NULL

2. Main Relationships

Relationship	Cardinality	Meaning
Patients → MedicalProfiles	1 : 0..1	One patient may have one optional medical profile.
Patients → EmergencyContacts	1 : many	A patient can register multiple emergency contacts.
Patients ↔ ChronicDiseases	many : many	Resolved by PatientChronicDiseases.
Patients ↔ Allergies	many : many	Resolved by PatientAllergies.
Patients ↔ Medications	many : many	Resolved by PatientMedications.
Patients → EmergencyRequests	1 : many	A patient can create multiple emergency cases.
EmergencyRequests → AITriageResults	1 : many	Keep multiple AI triage runs if the case is re-evaluated.
EmergencyRequests → EmergencyReviewDecisions / CaseStatusHistory	1 : many	Store operator decisions and case status audit trail.
Hospitals → Departments → Doctors	1 : many at each step	Each department belongs to one hospital; each doctor has one primary department in this draft.
Departments → HospitalBeds	1 : many	Beds are grouped under a department.
EmergencyRequests → hospital/department/doctor assignments	1 : many	Assignment records preserve changes over time.
Hospitals → Ambulances	1 : many	Each ambulance is associated with a hospital.
Hospitals → Paramedics	1 : many	Each paramedic works with one hospital.
Hospitals → HospitalStaff	1 : many	Hospital staff see and handle requests of their own hospital only.
Ambulances → AmbulanceAssignments / LocationHistory / StatusHistory	1 : many	Track case assignments, GPS points, and status transitions.
Paramedics → AmbulanceAssignments	1 : many	The hospital sends an ambulance with a free paramedic; the paramedic gets the job.
EmergencyRequests → Notifications / DoctorNotes / PatientTreatmentHistory	1 : many	Cases may have multiple notifications, notes, and treatment events.
EmergencyContacts → Notifications	1 : many	Emergency contacts get a message when the request is sent and when the status changes.
AspNetUsers → Patients / Doctors / Paramedics / HospitalStaff / Notifications	Identity relationship	Identity manages authentication; custom tables hold role-specific data.

3. ASP.NET Core Identity & Implementation Notes

Identity tables are generated by ASP.NET Core Identity, not custom tables in this dictionary. Common tables include AspNetUsers, AspNetRoles, AspNetUserRoles, AspNetUserClaims, AspNetUserLogins, AspNetUserTokens, and AspNetRoleClaims.

Before implementation

• Pick one consistent key type for Patients.PatientID and use it in every referencing FK.
• Ensure AspNetUsers.Id and every UserID/OperatorID/RecipientUserID FK have matching SQL types.
• Use composite primary keys for PatientChronicDiseases, PatientAllergies, and PatientMedications.
• Add indexes for foreign keys and frequently queried fields such as CurrentStatus, CreatedAt, and GPS timestamps.
• Standardize status/priority values and validate confidence score range (0–1) and latitude/longitude. Case statuses: Submitted, Confirmed, Ambulance Assigned, On the Way, Arrived, Picked Up, Going to Hospital, At Hospital, Completed.
• Define explicit delete behavior; do not accidentally cascade-delete emergency audit history.

Decisions to confirm

• The schema assumes each doctor has one primary department. If doctors can belong to multiple departments, add a DoctorDepartments junction table.
• Department capacity and individual HospitalBeds are both represented; decide whether both are needed.
• Hospital staff close the case (UC-10), so the case stores ClosedByUserID instead of ClosedByOperatorID.
• Finalize the exact fields and constraints with the team before implementation.